

FINTECH PRODUCT ANALYTICS

Investment Journey Case Study

Analytics Instrumentation, Product Thinking & SQL Analysis for the Investment Onboarding Journey.

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DISCIPLINES

Event Instrumentation · Product Thinking · SQL Analysis

PROJECT TYPE

Fintech Wealth Management · Personal Portfolio Project

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Project Overview: This case study designs and evaluates analytics instrumentation for a digital wealth platform helping users build long-term wealth through recurring Systematic Investment Plans (SIPs). It establishes rigorous event tracking, tests drop-off hypotheses, defines value metrics, and derives SQL queries for conversion, completion time, and drop-off analysis.

PART A

Event Specification

Fintech Product Analytics Case Study

Analytics instrumentation for the investment onboarding journey — from app open to first successful investment unit credit.

12 events specified.

Instrumentation Guidelines:

- **User-Initiated Events:** Direct in-app interactions, taps, and explicit form submissions.
- **System-Confirmed Events:** Asynchronous milestones confirmed only upon success responses from external KYC APIs, banking partners, payment gateways, or RTAs.
- **Hard Constraint:** Maximum 12 events specified, prioritized to deliver maximum analytical leverage across the onboarding funnel.

EVENT 01 OF 12

app_open

TRIGGER

Fire the event when the application finishes loading and the home/welcome screen is displayed to the user.

PROPERTIES

- **app_version** → Did the new release reduce app opens?
- **device_platform** (Android/iOS) → Are Android users behaving differently from iPhone users?
- **app_open_source** (Direct, Notification, Referral) → Which acquisition channel brings quality users?
- **app_load_time_seconds** → How long does the app take to become usable after launch?

QUESTION IT ANSWERS

How many users opened the app, and where are they coming from?

IF USERS DROP HERE

Users may leave because the app takes too long to load, crashes during startup, encounters network issues, or the welcome screen fails to engage them.

IMPROVEMENT

I would recommend improving app startup performance and reducing loading time.

EVENT 02 OF 12

mobile_verified

TRIGGER

Fire the event when the mobile verification API returns a successful response after the user enters the correct OTP.

PROPERTIES

- **otp_attempts** → Are users struggling with OTP verification?
- **verification_time_seconds** → Is OTP delivery taking too long?

QUESTION IT ANSWERS

How many users successfully verified their mobile number?

IF USERS DROP HERE

Users may abandon the process because they do not receive the OTP, enter an incorrect OTP multiple times, experience network issues, or lose interest.

IMPROVEMENT

I would recommend showing clear error messages, allowing OTP resend after a short interval.

EVENT 03 OF 12

email_verified

TRIGGER

Fire the email_verified event when the email verification API returns a successful response after the user enters the correct OTP.

PROPERTIES

- **otp_attempts** → How many attempts does email OTP verification take?
- **verification_time_seconds** → Is email OTP delivery causing delays?
- **email_domain** → Are users from certain email domains (e.g., Gmail, Outlook) experiencing more verification issues?

QUESTION IT ANSWERS

How many users successfully verified their email address?

IF USERS DROP HERE

Users may abandon the process because they don't receive the email OTP, enter an incorrect OTP multiple times, check the wrong email account, or have their OTP go to spam.

IMPROVEMENT

I would recommend sending the OTP to the user's primary email, providing a clear 'Resend OTP' option, informing users to check spam, and showing helpful error messages to reduce email verification failures.

EVENT 04 OF 12

pan_verified

TRIGGER

Fire the pan_verified event when the PAN verification API returns a successful response confirming the user's PAN details.

PROPERTIES

- **verification_time_seconds** → How long does PAN verification take?
- **pan_verification_status** → How often does PAN verification fail or succeed?
- **verification_attempts** → How many attempts does PAN verification require?

QUESTION IT ANSWERS

How many users successfully completed PAN verification?

IF USERS DROP HERE

Users may abandon because PAN verification fails, they enter incorrect details, or external KYC APIs respond slowly, causing frustration and drop-off.

IMPROVEMENT

I would recommend providing real-time PAN validation feedback, clear error messages explaining why verification failed, and estimated wait times when external APIs are slow.

EVENT 05 OF 12

sip_created

TRIGGER

Fire the sip_created event when the SIP setup API returns a successful response confirming the SIP has been created.

PROPERTIES

- **sip_frequency** → What SIP frequencies (daily, weekly, monthly) do users prefer?
- **sip_amount** → What is the typical SIP amount selected by users?
- **scheme_type** → Which investment schemes (Stocks, Gold, FDs) are most popular?
- **completion_time_seconds** → How long does SIP setup take from start to finish?

QUESTION IT ANSWERS

How many users successfully created their SIP?

IF USERS DROP HERE

Users may abandon during SIP setup because they are unsure about the investment amount, confused about scheme allocation, or encounter errors during SIP configuration.

IMPROVEMENT

I would recommend providing SIP amount suggestions based on income, simplified scheme selection with clear descriptions, and a progress indicator to keep users engaged during the setup process.

EVENT 06 OF 12

bank_verified

TRIGGER

Fire the bank_verified event when the bank verification API returns a successful response confirming the user's bank account.

PROPERTIES

- **bank_name** → Which banks have lower verification success rates?
- **verification_time_seconds** → How long does bank verification take?
- **verification_method** → Which bank verification method (net banking, debit card) is used more?

QUESTION IT ANSWERS

How many users successfully verified their bank account?

IF USERS DROP HERE

Users may abandon because they hesitate to share bank details due to trust concerns, the bank verification API fails, or bank verification takes too long.

IMPROVEMENT

I would recommend adding trust signals (security badges, encryption notices), simplifying the bank verification process, providing clear instructions, and handling API failures gracefully with informative error messages.

EVENT 07 OF 12

mandate_completed

TRIGGER

Fire the `mandate_completed` event when the UPI mandate approval is confirmed by the payment gateway or banking partner — not when the user taps confirm, but when the external system confirms success.

PROPERTIES

- `mandate_approval_time_minutes` → How long does UPI mandate approval take?
- `upi_app_used` → Which UPI apps (GPay, PhonePe, Paytm) are users choosing?
- `mandate_status` → What percentage of mandate requests succeed vs. fail?

QUESTION IT ANSWERS

How many users successfully completed UPI mandate setup?

IF USERS DROP HERE

Users may fail to complete the mandate because they cancel the UPI request, encounter UPI app errors, or lose internet connectivity during the mandate approval process.

IMPROVEMENT

I would recommend providing real-time mandate status updates, clear instructions for completing the UPI mandate in the user's preferred UPI app, and a retry option if the mandate fails.

EVENT 08 OF 12

additional_details_saved

TRIGGER

Fire the additional_details_saved event when the user submits the additional personal details form.

PROPERTIES

- **completion_time_seconds** → How long does it take to complete additional details?
- **fields_filled** → How many fields are users completing vs. skipping?
- **validation_errors** → Are users encountering form validation issues?

QUESTION IT ANSWERS

How many users successfully completed the additional details step?

IF USERS DROP HERE

Users may abandon the additional details step because the form is too long, information is unavailable, validation errors occur, or they are concerned about sharing additional personal information.

IMPROVEMENT

I would recommend simplifying the form by asking only essential information, clearly highlighting mandatory fields, providing real-time validation, and showing a progress indicator to reduce user frustration.

EVENT 09 OF 12

investment_consent_verified

TRIGGER

Fire the investment_consent_verified event when the investment consent OTP is successfully verified and the API returns a successful response.

PROPERTIES

- **otp_verification_time_seconds** → How long does OTP verification take?
- **otp_attempts** → How many attempts are required for successful OTP verification?
- **otp_delivery_time_seconds** → Is OTP delivery causing delays?

QUESTION IT ANSWERS

How many users successfully completed the investment consent verification?

IF USERS DROP HERE

Users may abandon the process because they do not receive the OTP, enter an incorrect OTP, face OTP verification failures, or lose interest while waiting for the verification.

IMPROVEMENT

I would recommend improving OTP delivery speed, providing a resend OTP option with a countdown timer, displaying clear error messages, and informing users about the purpose of the investment consent to build confidence.

EVENT 10 OF 12

first_unit_credited

TRIGGER

Fire the first_unit_credited event when the Registrar and Transfer Agent (RTA) or investment partner confirms that the first investment units have been successfully credited to the user's account.

PROPERTIES

- **unit_credit_time_hours** → How long does it take for units to be credited after SIP creation?
- **investment_scheme** → Which schemes have faster or slower unit credit?
- **investment_amount** → Does the investment amount affect the credit timeline?

QUESTION IT ANSWERS

How many users successfully received their first investment units?

IF USERS DROP HERE

Users may not receive their first investment units because the payment is pending, the mandate or transaction failed, processing is delayed by external partners, or the investment order was rejected.

IMPROVEMENT

I would recommend providing users with real-time investment status updates, expected unit credit timelines, and proactive notifications so they know their investment is being processed and don't lose confidence while waiting.

EVENT 11 OF 12

nominee_added

TRIGGER

Fire the nominee_added event when the user submits the nominee details form.

PROPERTIES

- **nominee_added** → How many users choose to add a nominee?
- **relation_to_nominee** → Which relationships (Parent, Spouse, Sibling, etc.) are most common?
- **completion_time_seconds** → How long does it take to add a nominee?

QUESTION IT ANSWERS

How many users successfully added a nominee?

IF USERS DROP HERE

Users may skip or abandon this step because it is optional, they do not have nominee details available, they do not understand its importance, or they find the process time-consuming.

IMPROVEMENT

I would recommend clearly explaining the benefits of adding a nominee, keeping the process simple, and allowing users to skip this step and add a nominee later.

EVENT 12 OF 12

save_dashboard_viewed

TRIGGER

Fire the save_dashboard_viewed event when the Save Dashboard is successfully loaded after the user's first investment units are credited.

PROPERTIES

- **portfolio_value** → What is the user's initial portfolio value?
- **active_sip_count** → How many active SIPs does the user have?
- **dashboard_load_time_seconds** → How long does the dashboard take to load?

QUESTION IT ANSWERS

How many users successfully reached and viewed the Save Dashboard?

IF USERS DROP HERE

Users may leave because the dashboard loads slowly, the investment status is unclear, technical issues occur, or they do not immediately understand the information displayed.

IMPROVEMENT

I would recommend optimizing the dashboard loading speed, displaying a clear investment summary, and providing simple explanations of portfolio metrics so users can easily understand their investments and stay engaged.

EXCLUDED EVENTS

Excluded Events & Justification

Excluded Event	Reason for Exclusion
welcome_screen_viewed	Excluded because app_open already marks the beginning of the onboarding funnel, making this event largely redundant.
sip_setup_started	Excluded because sip_created captures the final business outcome. Tracking the intermediate setup step adds limited analytical value under the 12-event constraint.
bank_link_started	Excluded because bank_verified measures successful completion, making the intermediate linking step less valuable for business analysis.

SELECTION STRATEGY

Due to the maximum limit of 12 events, I prioritized completion/outcome events over view/start events because they provide greater business value, clearer funnel analysis, and more actionable insights for product improvements.

PART B

A Hypothesis I Could Be Wrong About

HYPOTHESIS

I believe the **bank verification stage** leaks the most users in the investment onboarding journey.

REASONING

- Users may hesitate to share their bank account details due to trust concerns.
- Many users may not have their bank details readily available during onboarding.
- Bank verification depends on external banking APIs, which can fail or respond slowly.
- Any verification failure at this stage can reduce user confidence and increase drop-off.

WHAT EVIDENCE WOULD PROVE ME WRONG?

I would consider this hypothesis incorrect if the data shows:

- Bank verification has a consistently high completion rate.
- Most users who start bank verification complete it successfully.
- Another stage in the funnel has a significantly higher drop-off rate than bank verification.

PART C

North Star Metric & Counter Metric

NORTH STAR METRIC

Number of users who receive their first investment units (`first_unit_credited`).

This is the North Star Metric because it represents users who successfully complete the entire investment onboarding journey and become actual investors. It reflects the core business outcome and real customer value.

COUNTER METRIC

Customer Support Ticket Rate related to the investment onboarding journey.

This metric ensures that while increasing the number of users receiving their first investment units, the platform does not create a confusing or frustrating user experience. If customer support requests related to the investment journey increase, it indicates that product changes are causing user problems even though the North Star Metric is improving. Tracking this metric helps the team improve successful investments without sacrificing customer experience.

PART D

SQL Analysis & Assumptions

D.1 — SQL APPROACH

The SQL queries analyze the investment journey using the provided synthetic events table, which records every `event_name`, `user_id`, and `event_ts`. Each query answers one of the required business questions: funnel progression, completion time, and the largest drop-off. The queries rely only on event-level data and standard aggregation, without needing to parse the properties JSON column.

D.2 — SQL QUERIES SUMMARY

Query	Purpose
Query 1	Generates the step-by-step funnel showing users reaching each stage and step-over-step conversion.
Query 2	Calculates the median completion time between <code>app_open</code> and <code>first_unit_credited</code> for users who completed the journey.
Query 3	Identifies the largest drop-off between two consecutive stages in the investment funnel.

The complete SQL implementation is provided separately in the accompanying .sql script.

D.3 — ASSUMPTIONS

- Users may trigger the same event multiple times. For time-based analysis, I assumed the first occurrence represents the user's progression through the funnel.
- The brief does not specify whether events can occur out of order. I assumed the investment journey follows the documented event sequence.
- The brief does not specify how to handle users who restart the journey. I counted each user only once per funnel stage using `COUNT(DISTINCT user_id)`.
- The brief does not specify whether repeated events represent duplicate records or legitimate user actions. I assumed repeated events are valid user behavior.
- Only users who reached `first_unit_credited` were included when calculating the median completion time.
- I assumed the required outputs can be derived using `event_name`, `user_id`, and `event_ts`, without requiring additional information from the properties JSON column.